

CHASE: A Corridor-Heuristic Charging Scheduler for Ground and Aerial Vehicles in Wireless Rechargeable Sensor Networks

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Abstract. Under dense deployment a ground wireless charging vehicle (WCV) is too slow to meet every sensor deadline, while a purely aerial charger is capped by its battery. Existing hybrid schemes fall back from Nearest Job First to Earliest Deadline First only once the former is declared infeasible, so nodes due inside that switch window are already lost. We present CHASE, which applies both rules inside every WCV leg: it anchors the most urgent target by deadline, then folds in only the pending nodes lying within a narrow corridor along that leg. A second mechanism recovers a UAV launched from a moving WCV by solving the rendezvous with a provably contractive fixed-point iteration. Over 30 seeds CHASE sustains 82.8% survival at 1000 nodes against 74.8% for CHCS, peaking at a 10.0-point margin; removing the corridor alone costs 19.8 points.

Keywords: wireless rechargeable sensor networks, hybrid charging, unmanned aerial vehicle, deadline-aware scheduling, bypass corridor

1 Introduction

Wireless sensor networks underpin large-scale monitoring in smart-city instrumentation, precision agriculture and industrial telemetry [2, 23], with city-wide deployments now spanning thousands of sensors [15]. Across these settings the battery is the dominant failure mode. Wireless power transfer reframes that constraint as a scheduling problem [9]: mobile chargers visit sensors and recharge them in place, turning the network into a wireless rechargeable sensor network (WRSN). Xie et al. [20] showed that such a network can run indefinitely provided the charger’s cycle time stays within the slowest sensor’s battery lifetime.

That guarantee weakens as networks grow. Ground chargers travel at 1–5 m/s, which is adequate for a sparse field but too slow once sensors are dense or traffic spikes shorten deadlines [12]. Running two or three chargers in parallel [10, 14] does not remove the structural cost: every charger pays a long return-to-base trip when depleted, idling that share of the fleet during peak demand.

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A multi-node charging analysis shows the required charger count grows faster than network size [19], and He et al. [6] establish that periodic schedules cannot sustain the network once demand outstrips supply. This paper targets the regime where those bounds bind.

Work on this problem falls into three strands, reviewed in Section 2: ground-side schedulers, bounded by vehicle speed; UAV-side schedulers, bounded by onboard battery; and hybrid schemes that pair the two vehicle classes, which a recent survey identifies as the present frontier [8]. Two gaps persist. First, Earliest Deadline First (EDF) [13] and Nearest Job First (NJF) are still treated as alternatives rather than combined at fine granularity. Hybrid schemes that switch from NJF to EDF only after NJF declares infeasibility lose exactly those nodes whose deadline arrives during the switch [12]. Second, existing hybrid schemes such as CHCS [4] compute the point at which a WCV releases a carried UAV in closed form, but describe its recovery only as the UAV waiting for the moving vehicle to pass by again.

We adopt the three-ring service architecture of CHCS [4]—an inner zone served by base-station-anchored UAVs, a middle zone served by the WCV directly, and an outer set handed to the UAVs the WCV carries—and close both gaps. **CHASE** (Corridor-Heuristic Anchored-Schedule for EDF) combines EDF and NJF within every leg: each leg anchors an EDF-urgent target and augments the trip with on-the-way detours falling inside a narrow corridor along the leg. It degenerates to pure EDF when no detour fits and to pure NJF when all targets are equally urgent. The second contribution launches a WCV-carried UAV from the moving vehicle and recovers it by predicting the rendezvous point with a fixed-point iteration that we prove contractive whenever the UAV is faster than the WCV. Across 3,276 simulation runs we find the survival advantage to be large and statistically significant in the stressed regime, absent where the methods jointly saturate, and—by ablating the corridor—attributable to the corridor mechanism itself rather than to the inherited architecture.

2 Related Work

Ground-based charger scheduling. The single-charger formulation established the theoretical footing: the optimal charging path is a shortest Hamiltonian cycle [20], and energy provisioning bounds what any schedule can achieve [6]. Extending to several vehicles turns the problem into joint deployment and ordering [10, 14], while temporal-spatial prediction sharpens deadline handling [12]. A recent energy-adaptive scheme reorders service by residual node energy [17]. All of these remain bounded by ground travel speed, which is the constraint the aerial strand removes.

UAV-assisted charging. Aerial chargers trade capacity for an order-of-magnitude speed advantage, and the literature jointly optimises trajectory, scheduling, power and coverage [18, 16, 3]. Recent multi-AAV work adds one-to-many charging and spatiotemporally balanced task allocation [22], but remains purely aerial:

there is no air-ground heterogeneity and therefore no rendezvous problem. Limited onboard battery is the structural weakness of every UAV-only design.

Hybrid schemes and what they leave open. Pairing the two vehicle classes is the current frontier [8, 1]. P2S [11] and later collaborative designs [5] formalise task allocation between high-capacity ground vehicles and high-mobility aerial ones. The scheme closest to ours is CHCS [4], which integrates WCVs, WCV-carried UAVs and independent UAVs over three zones. Its WCV schedule is NJF-primary: the tour is built by nearest-job ordering, checked for deadline feasibility, and only when that check fails are the k most urgent middle-zone nodes reordered by EDF and the remainder appended in NJF order. The release point of a carried UAV is obtained in closed form from the WCV trajectory, but the paper describes recovery only as the UAV waiting for the WCV to pass by again. Both openings are structural rather than parametric: the NJF/EDF choice is made once per tour instead of once per leg, and recovery is left to a waiting rule. Rendezvous is itself well studied for data collection, where a mobile sink meets static nodes [21, 19]; what differs here is that both endpoints move.

3 System Model and Problem Statement

3.1 Network and Energy Model

Sensors $S = \{s_1, \dots, s_n\}$ are deployed uniformly at random on a disk of radius R with a base station (BS) at the centre. Each sensor has battery capacity B and drains at a rate derived from the first-order radio energy model [7]: sensing at a constant power plus transmitting its own traffic and relaying the traffic of its subtree in a BFS gradient routing tree rooted at the BS. The relay term makes the drain rate strongly position-dependent, so nodes near the BS deplete fastest. A sensor whose residual energy falls to $\theta_c B$ issues a charging request carrying a deadline $\tau_d(s_i)$, the time at which it would reach zero at its current rate. When a node dies it is removed from the routing tree and its subtree is rerouted, which shifts the relay load onto its neighbours.

3.2 Charging Resources and Service Architecture

Charging capacity is heterogeneous and, importantly, plural: the field is served by m WCVs of speed v_{WCV} , each carrying k UAVs, plus b further UAVs anchored at the BS. At initialisation the field is partitioned into m clusters by azimuth, with the cut angles chosen so that the total consumption of each cluster is approximately equal, and WCV w_j is assigned to cluster j for the whole run.

Within a cluster, service follows the three-ring division of CHCS [4], illustrated in Fig. 1. The *inner* zone is the disk of radius $D_{\max}/2$ about the BS, fixed once at initialisation and served exclusively by the BS-anchored UAVs. The split of the remaining requests into *middle* and *outer* is not a static ring: in each round the k pending nodes farthest from the centroid of that round's

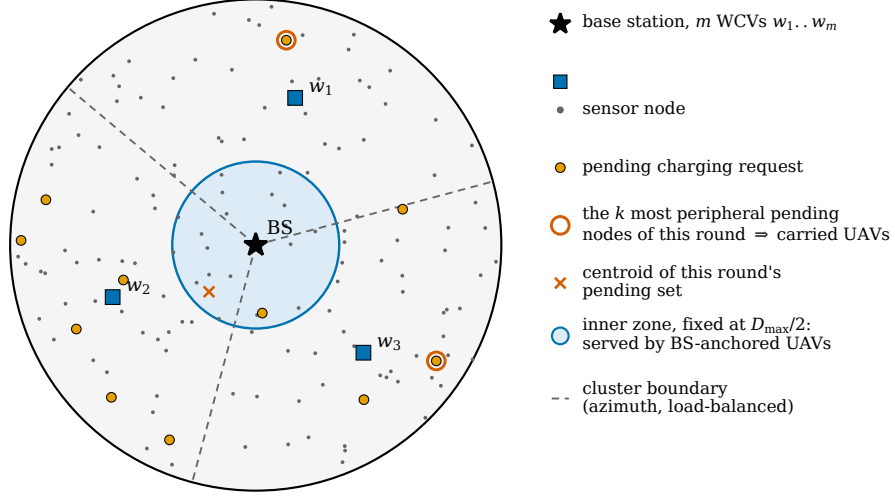


Fig. 1. Service architecture. The field is split by azimuth into m load-balanced clusters, one WCV each. The inner disk of radius $D_{\max}/2$ is fixed at initialisation and served by BS-anchored UAVs. Of the remaining pending requests, the k farthest from the pending set's centroid are handed to that WCV's carried UAVs each round; the rest are served by the WCV directly.

pending set are handed to the carried UAVs, and the rest are scheduled for the WCV itself. This distinction matters for the evaluation, because it means the middle/outer populations are a consequence of each scheduler's own decisions, whereas the inner population is purely geometric and identical across methods.

3.3 Problem Statement

Let $x_i \in \{0, 1\}$ indicate that sensor s_i is alive at the end of the horizon T_{end} , and let $A(s_i)$ be the vehicle assigned to serve s_i , arriving at $\tau_a(s_i)$. The scheduling problem is

$$\max \sum_{i=1}^n x_i \quad \text{subject to} \quad \tau_a(s_i) \leq \tau_d(s_i) \quad \forall s_i \in S_{\text{serve}}, \quad (1)$$

together with the physical constraints that each vehicle respects its own speed, that a carried UAV's outbound flight, hover charging and return flight fit inside its endurance, that every launched UAV is recovered, and that a WCV whose battery falls below a reserve returns to the BS to swap. A schedule that satisfies (1) for all pending nodes is called *available*; when no such schedule exists the scheduler must decide which nodes to give up, and it is that decision under overload that separates the methods evaluated in Section 5.

Table 1. Simulation parameters. Values are those recorded for the reported experiments; swept axes give the range, with the default in bold.

Parameter	Value	Parameter	Value
Field radius R	500 m	Sensor battery B	100 J
Sensors n	1000 (200–1400)	Request thresh. θ_c	0.3 (0.3–0.7)
WCVs m	3	Charging rate ρ	5 J s^{-1}
UAVs per WCV k	2	Sensing power	0.007 J s^{-1}
BS-anchored UAVs b	2	Transmission range	60 m
WCV speed v_{WCV}	5 m/s (2–10)	Corridor ratio c	0.3 (0.1–0.5)
UAV speed v_{UAV}	35 m/s	Rdv. tolerance ϵ	1 m
UAV range D_{max}	200 m	Rdv. iteration cap	10
WCV battery E_{WCV}	10 kJ (5–20)	Horizon T_{end}	10^5 s
WCV move power	10 J m^{-1}	Time step Δt	0.5 s
WCV reserve	$0.1 E_{\text{WCV}}$	UAV battery	300 J
UAV fly / hover power	10 / 1 J s^{-1}	Seeds per configuration	30

4 The CHASE Scheduler

4.1 Overall Flow

After a one-time initialisation (deployment, routing-tree construction, clustering by azimuth, and inner-zone fixing), the simulation advances in steps of Δt (Algorithm 1). Each round collects the current requests, partitions them per cluster, runs the CHASE schedule for that cluster’s WCV, plans the carried-UAV missions, dispatches the BS-anchored UAVs to the inner zone, then advances every vehicle and updates energy and liveness.

4.2 Anchoring and the Bypass Corridor

Pure EDF respects deadlines but produces long zigzag trajectories, because consecutive urgent nodes may sit far apart. Pure NJF travels efficiently but starves distant urgent nodes. CHASE keeps EDF as the backbone and recovers NJF’s efficiency only where it costs almost nothing: each leg is *anchored* at the most urgent pending node, and any node that is nearly on the straight path to that anchor is served en route (Algorithm 2).

Given a leg from $\mathbf{p}$ to $\mathbf{q}$, let $\ell = d(\mathbf{p}, \mathbf{q})$, $\vec{v} = \mathbf{q} - \mathbf{p}$, and let the corridor half-width be $\delta_{\text{max}} = c\ell$ for a corridor ratio c . A pending node s_i is a bypass candidate if and only if three conditions hold: its perpendicular distance to the line through $\mathbf{p}$ and $\mathbf{q}$ is at most δ_{max} (corridor); $\vec{v} \cdot (s_i - \mathbf{p}) > 0$, so it lies ahead of the current position (forward); and $d(\mathbf{p}, s_i) < \ell$, so it is no farther than the anchor itself (progress). Fig. 2 shows the admissible region the three conditions cut out, and one node failing each of them. Together they bound the detour: a candidate is accepted only if it is beside the path, ahead of the vehicle, and closer than the destination. Because the anchor is always appended last, the leg

Algorithm 1 Overall scheduling flow

Require: initialised network: sensors S , routing tree, clusters $1, \dots, m$

- 1: $t \leftarrow 0$
- 2: **while** $t < T_{\text{end}}$ **do**
- 3: $Q \leftarrow \{s \in S : E_s \leq \theta_c B\}$ ▷ pending requests
- 4: **for** each cluster $j = 1, \dots, m$ **do**
- 5: $Q_j \leftarrow Q \cap \text{cluster}_j$; remove the inner-zone members of Q_j
- 6: $O \leftarrow$ the k members of Q_j farthest from the centroid of Q_j
- 7: $(\sigma_w, \sigma_u) \leftarrow \text{CHASE}(w_j, Q_j \setminus O, O)$ ▷ Algorithm 2
- 8: plan a carried-UAV mission for each target in σ_u (Section 4.3)
- 9: **end for**
- 10: dispatch BS-anchored UAVs to the inner-zone requests
- 11: advance all vehicles by Δt ; deliver charge at rate ρ ; update energy
- 12: mark zero-energy nodes dead, remove them from the tree and reroute their subtrees
- 13: any WCV below its reserve returns to the BS and swaps its battery
- 14: $t \leftarrow t + \Delta t$
- 15: **end while**

still ends at the most urgent node, so folding in bypass nodes can only delay it by the detour they add, never displace it.

Each iteration removes at least the anchor, so the loop runs at most $|M|$ times and the candidate scan is $O(|M|)$, giving $O(|M|^2)$ per cluster per round in the worst case and markedly less in practice because each iteration also consumes its bypass set. The corridor ratio c is the method's only tuning parameter; we fix $c = 0.3$ and report its sensitivity in Section 5.6.

4.3 UAV Dispatch and Rendezvous around a Moving WCV

Because the WCV never stops for the outer targets, a UAV cannot take off from a fixed base. Each mission selects the carried UAV with the earliest free time, queries a piecewise-linear timeline of the WCV path for the take-off position $\mathbf{a}$, and computes the release point from $\mathbf{a}$, the next waypoint, the target s_o and its deadline. Three cases arise (Fig. 3a): the planned path already passes within D_{max} of the target, so the UAV is released as the WCV enters that disk; the path passes too far but the deadline still allows a detour, in which case the release point is the first root in $[0, 1]$ of the quadratic $\|\mathbf{e}(t) - s_o.\mathbf{p}\|^2 = D_{\text{max}}^2$ along the current segment; or the deadline is too near for either, and the UAV is released immediately.

Recovery is the harder half, and it is circular: where the UAV should fly back to depends on when it finishes charging, yet that time depends on how far the rendezvous is from the sensor. We break the loop by successive approximation (Fig. 3b). Starting from the rendezvous the UAV would reach if it simply retraced its outbound path, iteration k estimates the return time $t^{(k)} = t_0 + d(s_o.\mathbf{p}, \mathbf{a}^{(k-1)})/v_{\text{UAV}}$ and takes the WCV position at $t^{(k)}$ as the new rendezvous. Iteration stops when two successive points differ by less than

Algorithm 2 CHASE: anchored scheduling with corridor bypass**Require:** WCV w at $\mathbf{p}_w$, middle candidates M , outer candidates O **Ensure:** WCV schedule σ_w , carried-UAV schedule σ_u

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1:  $\sigma_w \leftarrow \langle \rangle$ ;  $\mathbf{p} \leftarrow \mathbf{p}_w$ 
2: while  $M \neq \emptyset$  do
3:    $s^* \leftarrow \arg \min_{s \in M} \tau_d(s)$ ;  $M \leftarrow M \setminus \{s^*\}$  ▷ EDF anchor
4:    $\ell \leftarrow d(\mathbf{p}, s^*. \mathbf{p})$ ;  $\delta_{\max} \leftarrow c\ell$ ;  $\vec{v} \leftarrow s^*. \mathbf{p} - \mathbf{p}$ 
5:    $B \leftarrow \{s \in M : \text{dist}(s. \mathbf{p}, \mathbf{p} s^*. \mathbf{p}) \leq \delta_{\max}$  ▷ corridor
      $\wedge \vec{v} \cdot (s. \mathbf{p} - \mathbf{p}) > 0 \wedge d(\mathbf{p}, s. \mathbf{p}) < \ell\}$  ▷ forward, progress
6:   sort  $B$  ascending by  $d(\mathbf{p}, s. \mathbf{p})$  ▷ NJF within the leg
7:    $\sigma_w \leftarrow \sigma_w \cdot B \cdot \langle s^* \rangle$ ;  $M \leftarrow M \setminus B$ ;  $\mathbf{p} \leftarrow s^*. \mathbf{p}$ 
8: end while
9:  $\sigma_u \leftarrow \text{sort } O \text{ ascending by } d(\mathbf{p}_w, s. \mathbf{p})$ 
10: return  $\sigma_w, \sigma_u$ 

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ϵ , or at the iteration cap. Should it not converge, the planner falls back to the take-off point $\mathbf{a}$ itself, which lies on the WCV path by construction, so the UAV can never be stranded off the trajectory.

Lemma 1 (Contractive rendezvous). *Let Γ be the WCV trajectory under linear interpolation, and let $T : \Gamma \rightarrow \Gamma$ map a candidate rendezvous to the next iterate. If $v_{\text{WCV}} < v_{\text{UAV}}$, then T is Lipschitz with constant $L = v_{\text{WCV}}/v_{\text{UAV}} < 1$, hence a contraction.*

Proof. Linear interpolation makes $\text{WCV}(t)$ v_{WCV} -Lipschitz in t , on travel segments at speed v_{WCV} , on charging segments at speed 0, and across joints by continuity. For two candidates the return times satisfy $|t_1^{(k)} - t_2^{(k)}| \leq \|\mathbf{a}'_1 - \mathbf{a}'_2\|/v_{\text{UAV}}$ by the reverse triangle inequality, so

$$\|T(\mathbf{a}'_1) - T(\mathbf{a}'_2)\| \leq v_{\text{WCV}}|t_1^{(k)} - t_2^{(k)}| \leq \frac{v_{\text{WCV}}}{v_{\text{UAV}}} \|\mathbf{a}'_1 - \mathbf{a}'_2\|. \quad (2)$$

Since Γ is a closed subset of $\mathbb{R}^2$ and T maps it into itself, the Banach fixed-point theorem gives a unique rendezvous with geometric convergence at rate L^k . At the configuration of Table 1, $L = 5/35 \approx 0.14$, so a ten-iteration cap is ample and the fallback is a formality rather than an operating mode.

5 Evaluation

5.1 Setup and Compared Methods

We use a discrete-time simulator with the parameters of Table 1. Node density is swept from 200 to 1400 sensors; WCV battery capacity, WCV speed, node energy threshold and corridor ratio are swept separately around the default operating point. Every configuration is run over the same 30 fixed seeds, so the five methods see identical network instances and every comparison below is *paired*: we report the mean of the per-seed difference, its standard error, and a

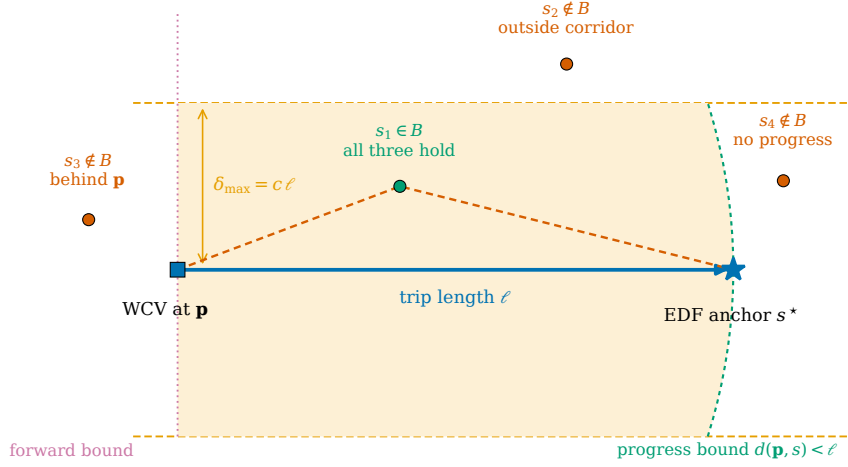


Fig. 2. Bypass admissibility around one leg. The shaded region is the intersection of the three conditions: inside the corridor of half-width $\delta_{\max} = c\ell$, ahead of $\mathbf{p}$, and nearer to $\mathbf{p}$ than the anchor. Only s_1 satisfies all three and is folded into the trip; s_2 , s_3 and s_4 each fail exactly one.

two-sided Wilcoxon signed-rank test over the 30 pairs. All methods share the same simulator core, the same clustering, the same inner-zone service and the same middle/outer split rule, so the only difference between them is the ordering rule applied to a cluster's middle-zone requests:

- **CHCS** [4] is the primary comparison method: NJF-primary with EDF rescue on infeasibility, as described in Section 2. This is our reimplementation of the published rule from the paper's description, not the authors' code, so it should be read as a faithful implementation of the rule rather than a calibrated reproduction of the original results.
- **EDF** orders the middle-zone requests purely by deadline, and skips any node it computes it cannot reach in time. It is not merely a sanity baseline: because it shares every other component with CHASE, it *is* CHASE with the corridor removed, and we use it as such in Section 5.3. It follows the classical earliest-deadline rule [13] rather than reproducing any particular WRSN scheduler.
- **GREEDY** orders by an efficiency score, urgency divided by distance to the vehicle. **FCFS** orders by the time a node entered the pending queue. Neither reproduces a published scheduler; they are trivial ordering rules included to confirm that the comparison does not collapse to an obviously unreasonable ordering.

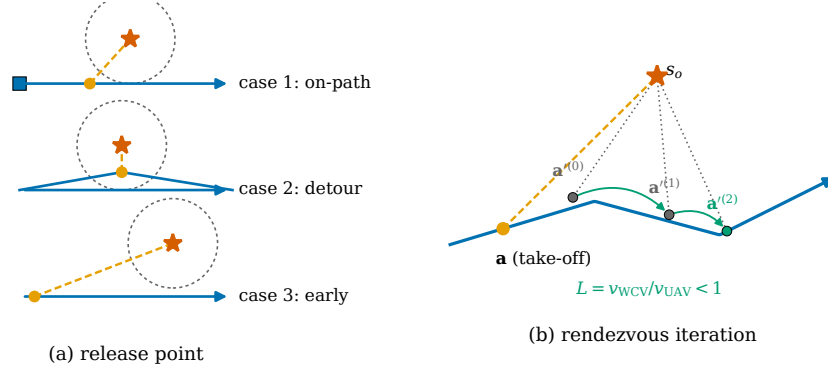


Fig. 3. (a) The three release cases: the WCV path enters the target's $D_{\max}$ disk, a detour is inserted to make it do so, or the deadline forces immediate release. (b) Recovery: successive iterates $\mathbf{a}^{(0)}, \mathbf{a}^{(1)}, \dots$ walk along the WCV path to the rendezvous, contracting by $L = v_{\text{WCV}}/v_{\text{UAV}}$ per step (Lemma 1).

5.2 Survival across Node Density

Table 2 and Fig. 4 report survival against node density. Up to $n = 400$ every method holds the ceiling: charging supply exceeds demand and ordering barely matters. From $n = 600$ the three trivial baselines break away first, dropping to 88–93% while CHASE and CHCS are still at 99.9–100%. The two deadline-aware methods separate only from $n = 1000$.

We state the significance boundary explicitly, because it is where the compressed narrative is easiest to overstate. **At $n = 600$ and $n = 800$ CHASE and CHCS are not distinguishable:** the paired differences are -0.13 and $+0.95$ points with paired standard errors of 0.11 and 0.86 , and Wilcoxon $p = 0.109$ and $p = 0.325$. At $n = 600$ CHCS is in fact nominally ahead. From $n = 1000$ the gap becomes large and unambiguous: $+7.97$ points ($p < 10^{-7}$), peaking at $+9.99$ points at $n = 1200$ and $+9.09$ at $n = 1400$. Against the weakest method the margin at the default point is 21.5 points. The ordering of the five methods is stable over the whole divergent regime $n \geq 800$.

5.3 What the Corridor Contributes

Because EDF differs from CHASE only in the removal of the corridor, the CHASE–EDF gap isolates the contribution of the proposed mechanism rather than of the inherited architecture. That gap is $+11.66 \pm 0.49$ points at $n = 600$, rises to $+21.97 \pm 0.40$ at $n = 800$, and is $+19.81 \pm 0.34$, $+17.38 \pm 0.23$ and $+15.33 \pm 0.24$ at $n = 1000, 1200$ and 1400 (all $p < 10^{-7}$). The corridor is therefore doing most of the work: the inherited three-ring architecture is worth whatever EDF achieves, and the corridor adds about twice again the margin that separates CHASE from CHCS.

Table 2. Survival rate (%) versus node density, 30-seed mean. The last two columns give the paired CHASE–CHCS difference with its paired standard error, and the two-sided Wilcoxon signed-rank p over the 30 seeds. At $n \leq 400$ the two methods are identical on every seed, so no test is defined.

n	CHASE	CHCS	GREEDY	EDF	FCFS	Δ vs CHCS	p
200	100.00	100.00	100.00	100.00	100.00	± 0.00	—
400	100.00	100.00	99.66	99.24	99.27	± 0.00	—
600	99.87	100.00	93.13	88.21	87.76	-0.13 ± 0.11	0.109
800	95.48	94.52	79.22	73.51	72.01	$+0.95 \pm 0.86$	0.325
1000	82.76	74.79	67.32	62.95	61.27	$+7.97 \pm 0.91$	$< 10^{-7}$
1200	73.71	63.72	59.90	56.33	54.73	$+9.99 \pm 0.59$	$< 10^{-7}$
1400	66.35	57.26	54.30	51.02	49.64	$+9.09 \pm 0.61$	$< 10^{-7}$

Two caveats belong with this ablation, and they pull in opposite directions. EDF additionally skips nodes it computes it cannot reach before their deadline, which CHASE does not do; that filter can only help EDF, so the measured gap understates the corridor’s effect. Against that, the corridor is not free: at $n = 1000$ CHASE drives 362.8 km more than EDF, a 47.8% increase in ground mileage, and the increase is 35–45% across $n \geq 800$. Relative to its own ablation, then, CHASE clearly buys survival with travel. The comparison in which it does not is the one against CHCS, and Section 5.5 explains why the two comparisons point in opposite directions.

5.4 Where the Gain Lands

Decomposing survival by zone at the default configuration shows that the improvement falls where the treatment applies. All five methods keep the inner zone at exactly 100%: those 201.5 nodes on average are served by the BS-anchored UAVs, which no scheduler under test modifies. The inner zone is a sanity check, not a control. Outside it, CHASE keeps 78.41% of the 798.5 remaining nodes alive against 68.43% for CHCS, 59.07% for GREEDY, 53.60% for EDF and 51.49% for FCFS. The aggregate 7.97-point margin is thus a diluted view of a 9.98-point effect concentrated in the region the WCVs and their carried UAVs actually serve.

We do not split the non-inner nodes further into middle and outer for cross-method comparison: as Section 3 notes, each method assigns nodes to those two groups through its own scheduling decisions, so per-group rates would compare different node populations.

5.5 Cost and the Vehicle Split

Against CHCS the sign of the travel effect changes with load (Fig. 5a). At low density CHASE drives *more*: by 12.3 ± 1.6 km at $n = 200$, 23.1 ± 2.1 at $n = 400$ and 44.5 ± 3.0 at $n = 600$, all significant, because with few pending

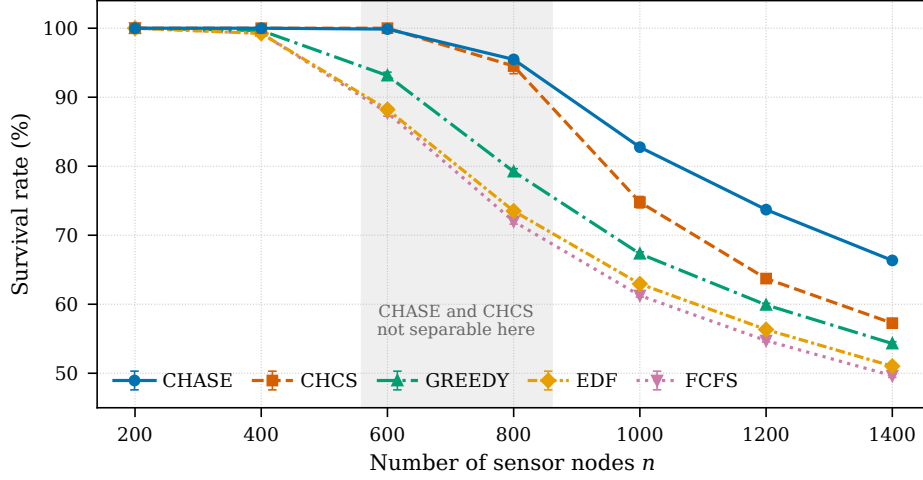


Fig. 4. Survival rate versus node density, 30-seed mean with ± 1 standard error. The shaded band marks the two densities at which CHASE and CHCS cannot be separated (Table 2). The gap between CHASE and EDF is the contribution of the corridor alone, since the two share every other component (Section 5.3).

nodes the corridor rarely finds a passenger and the EDF backbone pays its zigzag cost. From $n = 1000$ the ordering reverses and widens: -99.4 ± 23.1 km, then -257.6 ± 23.3 and -354.2 ± 20.4 , so at $n = 1400$ CHASE drives 899 km against CHCS's 1254 km, a 28% reduction. The summary is therefore not that CHASE travels less, but that its ground mileage saturates and then falls with load while CHCS's does not.

Fig. 5b identifies what changes. All five methods hand at most $k = 2$ nodes per round to the carried UAVs, yet the resulting division of labour diverges sharply. CHASE completes 82–89% of all charging events with the carried UAVs at every density, as do the three trivial baselines. CHCS matches that up to $n = 600$ and then collapses: 62.1% at $n = 800$, 17.4% at $n = 1000$, and 3.6% at $n = 1400$. Two further measurements are consistent with the same picture. At $n = 1000$ CHCS's feasibility check fails and hands over to its rescue path 16,728 times per run, roughly once every six seconds of simulated time, so the rescue is its operating mode rather than an exception; and its vehicles complete only 1,629 service cycles over the horizon against CHASE's 4,638, with a mean wait from request to charge of 905s against 192s. Read together these say that under overload the rescue path loads the whole reachable middle set onto the slow ground vehicle in one long cycle, which both stretches the cycle and starves the fast carried UAVs of dispatch opportunities. This is the mechanism the measurements are consistent with, not an independent experiment, and it is a property of our reimplementations of the CHCS rule.

The same reversal shows in a load-normalised cost. Taking total vehicle energy per surviving node, CHASE falls from 11.46 kJ at $n = 200$ to 6.48 kJ at

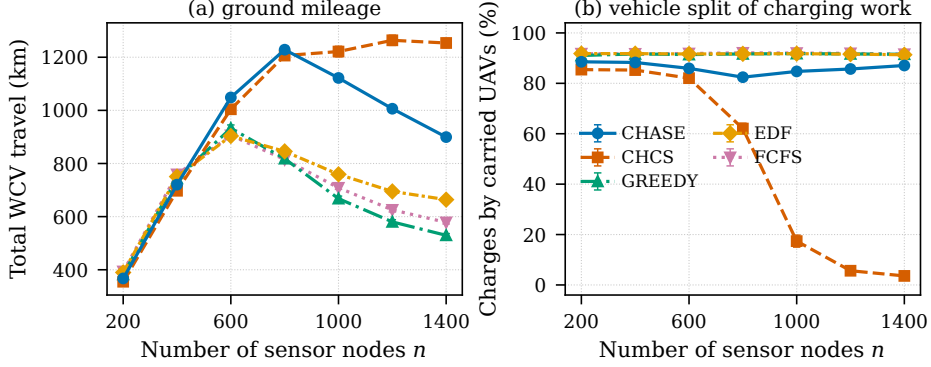


Fig. 5. (a) Total WCV travel and (b) the share of charging events completed by the carried UAVs, versus node density, 30-seed mean with ± 1 standard error. CHASE keeps the aerial share between 82 and 89% at every density; CHCS’s collapses from 85.4% to 3.6%, which is the same load range over which its ground mileage keeps climbing in (a).

$n = 1400$, as do the three baselines, while CHCS alone rises after $n = 800$, reaching 12.43 kJ at $n = 1000$ against CHASE’s 8.75 kJ. We note for completeness that the trivial baselines sit *below* CHASE on this metric (6.75–7.92 kJ at $n = 1000$), which is not efficiency: they abandon over 15 points more of the network, so their energy is amortised over fewer survivors.

5.6 Other Operating Points

Table 3 sweeps the remaining axes. The pattern is consistent and it bounds the claim: CHASE’s advantage is a stressed-regime advantage. It leads CHCS by 8.6–9.0 points across the whole battery range, by 9.1 and 9.5 points at the two lowest WCV speeds, and by 7.7 points at the tightest request threshold. Where the operating point is slack, the advantage disappears and in places reverses by a small but measurable amount: at $v_{WCV} = 10$ m/s the two are indistinguishable (-0.27 ± 0.61), and at request thresholds of 0.5 and above CHCS reaches exactly 100% while CHASE sits at 98.8–99.8%, a deficit of 0.2–1.2 points that is small but statistically real. Both regimes are ones in which the network is essentially saturated and any deadline-aware ordering suffices; we report them rather than restrict the sweeps to where CHASE wins.

Sweeping the corridor ratio c over $\{0.1, \dots, 0.5\}$ moves CHASE’s survival by 0.97 points in total, against a seed-to-seed standard deviation of about 2.2 points at the same operating point: the parameter is not worth tuning per scenario. At high node density the corridor almost always contains some candidate, so widening it saturates quickly. One caveat belongs with this result: the corridor sweep was produced by a later re-run of the simulator than the sweeps above, and that re-run sits about 10 points higher in absolute survival at the nominal default point. The five corridor values are internally consistent with each other,

Table 3. CHASE survival and the paired CHASE–CHCS difference in points (paired standard error in parentheses) across the remaining sweeps, $n = 1000$, 30 seeds each. The 5 kJ battery point is omitted: only 6 of its 30 CHASE runs survive in the archived data, so it is not comparable with the rest.

<i>WCV battery</i> (kJ)					
sweep value	10	15	20		
CHASE (%)	83.03	82.60	82.30		
Δ vs CHCS	+8.71 (0.68)	+8.58 (0.75)	+9.02 (0.74)		
<i>WCV speed</i> (m/s)					
sweep value	2	4	6	8	10
CHASE (%)	59.57	76.90	87.03	93.30	97.05
Δ vs CHCS	+9.11 (0.42)	+9.49 (0.62)	+5.54 (0.72)	+2.44 (0.82)	−0.27 (0.61)
<i>Request threshold θ_c</i>					
sweep value	0.3	0.4	0.5	0.6	0.7
CHASE (%)	83.60	94.99	99.21	99.81	98.78
Δ vs CHCS	+7.71 (0.81)	−0.18 (1.14)	−0.79 (0.24)	−0.19 (0.08)	−1.22 (0.24)

so the insensitivity conclusion holds, but their absolute level is not comparable with Table 2 and we do not quote it.

5.7 Threats to Validity

The evaluation is simulation-only, so localisation error, terrain and wind are absent. CHCS is our reimplementaion, and the two openings we identify are ones we read in its description rather than in its code. The three trivial baselines were run as a separate batch several weeks after the CHASE, CHCS and EDF runs; the CHASE–CHCS and CHASE–EDF comparisons on which every claim above rests are same-batch and same-seed, but the margins over GREEDY and FCFS cross batches. Finally, the archived simulator configuration has drifted since these runs were produced, so reproduction should use the configuration snapshot recorded with the results rather than the current default.

6 Conclusion

CHASE unifies EDF and NJF inside a single WCV leg through a bypass corridor, and recovers UAVs launched from a moving WCV by a provably contractive rendezvous iteration. Its advantage is a stressed-regime advantage, and the evaluation places the boundary precisely: from $n = 1000$ upward it leads CHCS by 8.0 to 10.0 points with $p < 10^{-7}$, at low WCV speeds by about 9 points, and at the tightest request threshold by 7.7 points; at $n \leq 800$, at 10 m/s, and at relaxed request thresholds the two methods are indistinguishable or CHCS is marginally ahead. Ablating the corridor while holding every other component fixed costs 15 to 22 points wherever the network is under load ($n \geq 800$), and

11.7 points even at $n = 600$, which locates the gain in the proposed mechanism rather than in the inherited architecture.

Three limitations bound these claims. The corridor buys survival with ground mileage relative to its own ablation, and is cheaper than CHCS only beyond $n = 1000$, where CHCS's own ground load is climbing. The zone partition is inherited rather than optimised, and a partition adapted to observed demand may do better. The contraction guarantee assumes the UAV is strictly faster than the WCV; the fallback covers the remaining case but does so conservatively. Validating the rendezvous predictor on a physical platform is the most direct next step.

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